

QM II 2024

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Question 1

0.0.1 (a) (i)

We have $l_a = 2$ and $l_b = 1$ thus as we know $|m_i| \leq l_i$ we have $m_A = -2, -1, 0, 1, 2$ and $m_B = -1, 0, 1$

$$l_{tot} \in [|l_A - l_B|, l_A + l_B] = [1, 3] = 1, 2, 3 \quad (1)$$

$$m_{tot} \in [-l_{tot}, +l_{tot}] = [-3, 3] = -3, -2, -1, 0, 1, 2, 3 \quad (2)$$

$$(3)$$

(ii)

Writing $|3, 3\rangle$ in $|l_A, m_A; l_B, m_B\rangle$

$$|3, 3\rangle \rightarrow |2, 2; 1, 1\rangle \quad (4)$$

where the we are give l_a and l_B but m_A and m_B must add to m_{tot} and are maximal here.

$$\hat{L}^-|3, 3\rangle = \hbar\sqrt{3(3+1) - 3(3-1)}|3, 3-1\rangle = \hbar\sqrt{6}|3, 2\rangle \quad (5)$$

$$\hat{L}^-|2, 2; 1, 1\rangle = (\hat{L}_A^- + \hat{L}_B^-)|2, 2\rangle \otimes |1, 1\rangle \quad (6)$$

$$= (\hat{L}_A|2, 2\rangle) \otimes |1, 1\rangle + |2, 2\rangle \otimes (\hat{L}_B^-|1, 1\rangle) \quad (7)$$

$$= (\hbar\sqrt{6-2}|2, 1\rangle) \otimes |1, 1\rangle + |2, 2\rangle \otimes (\hbar\sqrt{2}|1, 0\rangle) \quad (8)$$

$$\hat{L}^-|3, 3\rangle = 2\hbar|2, 1; 1, 1\rangle + \sqrt{2}\hbar|2, 2; 1, 0\rangle \quad (9)$$

Combining we have

$$|3, 2\rangle = \sqrt{\frac{2}{3}}|2, 1; 1, 1\rangle + \sqrt{\frac{1}{3}}|2, 2; 1, 0\rangle \quad (10)$$

$$(11)$$

Now for $|3, 1\rangle$

$$\hat{L}^-|3, 2\rangle = \hbar\sqrt{12-2}|3, 1\rangle \quad (12)$$

$$\hat{L}^-|3, 2\rangle = \hbar\sqrt{10}|3, 1\rangle \quad (13)$$

$$\hat{L}^-|3, 2\rangle = \sqrt{\frac{2}{3}}\hat{L}^-|2, 1; 1, 1\rangle + \sqrt{\frac{1}{3}}\hat{L}^-|2, 2; 1, 0\rangle \quad (14)$$

$$= \sqrt{\frac{2}{3}} \left[(\hat{L}_A|2, 1\rangle) \otimes |1, 1\rangle + |2, 1\rangle \otimes (\hat{L}_B^-|1, 1\rangle) \right] + \sqrt{\frac{1}{3}} \left[(\hat{L}_A^-|2, 2\rangle) \otimes |1, 0\rangle + |2, 2\rangle \otimes (\hat{L}_B^-|1, 0\rangle) \right] \quad (15)$$

$$= \sqrt{\frac{2}{3}} \left[\hbar\sqrt{6}|2, 0; 1, 1\rangle + \hbar\sqrt{2}|2, 1; 1, 0\rangle \right] + \sqrt{\frac{1}{3}} \left[\sqrt{5}\hbar|2, 1; 1, 0\rangle + \sqrt{2}\hbar|2, 2; 1, -1\rangle \right] \quad (16)$$

$$= \left[2\hbar|2, 0; 1, 1\rangle + \hbar\sqrt{\frac{4}{3}}|2, 1; 1, 0\rangle \right] + \left[\sqrt{\frac{5}{3}}\hbar|2, 1; 1, 0\rangle + \sqrt{\frac{2}{3}}\hbar|2, 2; 1, -1\rangle \right] \quad (17)$$

$$= [2\hbar|2, 0; 1, 1\rangle] + \left[\sqrt{\frac{16}{3}}\hbar|2, 1; 1, 0\rangle + \sqrt{\frac{2}{3}}\hbar|2, 2; 1, -1\rangle \right] \quad (18)$$

$$(19)$$

So we combine to get

$$|3, 1\rangle = \frac{1}{\sqrt{10}} \left[2|2, 0; 1, 1\rangle + \sqrt{\frac{2}{3}}|2, 2; 1, 0\rangle + \sqrt{\frac{16}{3}}\hbar|2, 1; 1, 0\rangle \right] \quad (20)$$

Finish 3,1 and do 3,0

(iii)